

CET 479

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Spring 2023

In class Activity – Feb 15, 2023.

LAB_04 : Create and change file and folder Linking

1. Login using your non-root user account.

\$ **login** Van : Log in user name is van

Note – In today's exercise, you can switch to root account if you get a warning "permission denied" or you can grant all permissions for time being using **chmod 777**.

```
localhost login: Van
Password:
Last failed login: Wed Feb 22 14:27:38 EST 2023 on tty1
There was 1 failed login attempt since the last successful login.
[Van@localhost ~]$
```

2. Go to your home directory and create a directory called **tmplinking**.

\$ **mkdir** TEMPLINKING : Create a directory name is **tmplinking**

```
localhost login: Van
Password:
Last failed login: Wed Feb 22 14:27:38 EST 2023 on tty1
There was 1 failed login attempt since the last successful login.
[Van@localhost ~]$ mkdir TEMPLINKING
[Van@localhost ~]$ _
```

- Now, go the **tmplinking** directory and create a file (you can choose the file name).

\$ **cd TEMPLINKING** : Switch to directory current is **tmplinking**

\$ **touch file_01.txt** : Create a file name is **linkingfile_01**

```
localhost login: Van
Password:
Last failed login: Wed Feb 22 14:27:38 EST 2023 on tty1
There was 1 failed login attempt since the last successful login.
[Van@localhost ~]$ mkdir TEMPLINKING
[Van@localhost ~]$ cd TEMPLINKING
[Van@localhost TEMPLINKING]$ touch file_01.txt
[Van@localhost TEMPLINKING]$ _
```

- Link the file to a second file using hard link concept. [**ln <existing_file> <new_file>**]

\$ **ln file_01.txt link_hardfile_01.txt** : create a hard
link file from linkingfile_01 to link_hardfile_01

```
localhost login: Van
Password:
Last failed login: Wed Feb 22 14:27:38 EST 2023 on tty1
There was 1 failed login attempt since the last successful login.
[Van@localhost ~]$ mkdir TEMPLINKING
[Van@localhost ~]$ cd TEMPLINKING
[Van@localhost TEMPLINKING]$ touch file_01.txt
[Van@localhost TEMPLINKING]$ ls -l
total 0
-rw-r--r--. 1 Van Van 0 Feb 22 14:39 file_01.txt
[Van@localhost TEMPLINKING]$ ln file_01.txt link_hardfile_01.txt
[Van@localhost TEMPLINKING]$
```

\$ **ls -i link_hardfile_01.txt** : Verify linking file with inode number of linking file. (# **9 494 042**)

```
[Van@localhost TMPLINKING]$ ls -i link_hardfile_01.txt
9494042 link_hardfile_01.txt
[Van@localhost TMPLINKING]$ _
```

6. Again, create a new file, and using the soft link concept taught in class, link another file to it. Use absolute path names with symbolic links if need be. [**ln -s <existingfile> <newfile>**]

\$ **ln -s file_01.txt link_softfile_02.txt** : create a soft link file from linkingfile_02 to link_softfile_02

\$ **ln -sf link_softfile_02.txt** : to create a symbolic link overwrites the destination file or directory if it already exists, without prompting the user for confirmation.

```
[Van@localhost TMPLINKING]$ ln file_01.txt link_softfile_02.txt
[Van@localhost TMPLINKING]$ ln -s file_01.txt link_softfile_02.txt
ln: failed to create symbolic link 'link_softfile_02.txt': File exists
[Van@localhost TMPLINKING]$ ln -sf file_01.txt link_softfile_02.txt
[Van@localhost TMPLINKING]$ _
```

7. Check if you can see a difference in inode number. What did you observe?

\$ **ls -i** : Verify linking file with inode (*short for index node*) number of linking file.

```
[Van@localhost TMPLINKING]$ ln file_01.txt link_softfile_02.txt
[Van@localhost TMPLINKING]$ ln -s file_01.txt link_softfile_02.txt
ln: failed to create symbolic link 'link_softfile_02.txt': File exists
[Van@localhost TMPLINKING]$ ln -sf file_01.txt link_softfile_02.txt
[Van@localhost TMPLINKING]$ ls -i link_softfile_02.txt
9494043 link_softfile_02.txt
[Van@localhost TMPLINKING]$
```

8. Check if the files are linked using [**ls -l**]

\$ **ls -l** : Verify linking file with inode number of linking file.

```
[Van@localhost TMPLINKING]$ ls -i link_softfile_02.txt
9494043 link_softfile_02.txt
[Van@localhost TMPLINKING]$ ls -l link_softfile_02.txt
lrwxrwxrwx. 1 Van Van 11 Feb 22 14:49 link_softfile_02.txt -> file_01.txt
[Van@localhost TMPLINKING]$
```

9. Repeat the soft link concept with two directories. What did you observe?

\$ **mkdir DIR_01** : Create 1 directory name is **DIR_01**

\$ **mkdir DIR_02** : Create 1 directory name is **DIR_02**

\$ ln -s DIR_01 LINK_DIR_01 : Create 1 Linking directory name is LINK_DIR_01

\$ ln -s DIR_02 LINK_DIR_02 : Create 1 Linking directory name is LINK_DIR_02

```
[Uan@localhost TMPLINKING]$ mkdir DIR_01
[Uan@localhost TMPLINKING]$ mkdir DIR_02
[Uan@localhost TMPLINKING]$ ln -s DIR_01 LINK_DIR_01
[Uan@localhost TMPLINKING]$ ln -s DIR_02 LINK_DIR_02
[Uan@localhost TMPLINKING]$ tree
.
├── DIR_01
├── DIR_02
├── file_01.txt
├── LINK_DIR_01 -> DIR_01
├── LINK_DIR_02 -> DIR_02
├── link_hardfile_01.txt
└── link_softfile_02.txt -> file_01.txt
```

\$ ls -i LINK_DIR_02 : Verify Directories created

```
.
├── DIR_01
├── DIR_02
├── file_01.txt
├── LINK_DIR_01 -> DIR_01
├── LINK_DIR_02 -> DIR_02
├── link_hardfile_01.txt
└── link_softfile_02.txt -> file_01.txt

4 directories, 3 files
[Uan@localhost TMPLINKING]$ ls -i LINK_DIR_02
[Uan@localhost TMPLINKING]$ ls -l LINK_DIR_02
lrwxrwxrwx. 1 Uan Uan 6 Feb 22 14:57 LINK_DIR_02 -> DIR_02
[Uan@localhost TMPLINKING]$ _
```

10. Remove the files and directory.

To use either of these commands.

- [For to remove a hard link :](#)

First identify the name of file by command [`ls -l < file name>`].

```
$ ls -l link_hardfile_01.txt      : Original file
ls
```

Next use Command `rm` or `unlink` for delete link-hard file

```
$ rm link_hardfile_01.txt        : Delete a linking file
```

```
$ unlink link_hardfile_01.txt    : Delete a linking file
```

```
[Van@localhost TMPLINKING]$ ls link_hardfile_01.txt
link_hardfile_01.txt
[Van@localhost TMPLINKING]$ rm link_hardfile_01.txt
[Van@localhost TMPLINKING]$ ls link_hardfile_01.txt
ls: cannot access 'link_hardfile_01.txt': No such file or directory
[Van@localhost TMPLINKING]$ ls -i link_hardfile_01.txt
ls: cannot access 'link_hardfile_01.txt': No such file or directory
[Van@localhost TMPLINKING]$ tree
.
├── DIR_01
├── DIR_02
├── file_01.txt
├── LINK_DIR_01 -> DIR_01
├── LINK_DIR_02 -> DIR_02
└── link_softfile_02.txt -> file_01.txt

4 directories, 2 files
[Van@localhost TMPLINKING]$ _
```

- [For to remove a symbolic link](#) :

\$ **rm -i link_softfile_02.txt** : Delete a symbolic linking file

\$ **rm -f link_softfile_02.txt** : Delete a linking file without confirm

\$ **rm -rf TMPLINKING** : Delete folder with all files linking in folder

```

— DIR_01
— DIR_02
— file_01.txt
— LINK_DIR_01 -> DIR_01
— LINK_DIR_02 -> DIR_02
— link_softfile_02.txt -> file_01.txt

2 directories, 2 files
[Van@localhost TMPLINKING]$ rm -rf LINK_DIR_01
[Van@localhost TMPLINKING]$ rm -rf LINK_DIR_02
[Van@localhost TMPLINKING]$ tree

— DIR_01
— DIR_02
— file_01.txt
— link_softfile_02.txt -> file_01.txt

2 directories, 2 files
[Van@localhost TMPLINKING]$ _

```

11. Logout.

\$ **logout**

```

.
— DIR_01
— DIR_02
— file_01.txt
— link_softfile_02.txt -> file_01.txt

2 directories, 2 files
[Van@localhost TMPLINKING]$ logout_

```

